



Tensile and low cycle fatigue properties of EUROFER97-steel after 16.3 dpa neutron irradiation at 523, 623 and 723 K



E. Materna-Morris*, A. Möslang, H.-C. Schneider

Karlsruhe Institute of Technology KIT, Institute for Applied Materials IAM, P.O. Box 3640, D-76021 Karlsruhe, Germany

ARTICLE INFO

Article history:

Available online 27 March 2013

ABSTRACT

Neutron-irradiated specimens of the reduced-activation tempered martensitic steel EUROFER97 were tested by tensile and low cycle conditions to detect the impact of irradiation on strength and lifetime. The irradiation temperature ranged from 523 to 723 K with an accumulated dose of up to 16.3 dpa. Tensile tests revealed a significant irradiation-induced hardening below 673 K with a peak of ~430 MPa at 573 K but none was seen at 723 K, as expected. Despite the significant irradiation-induced reduction of uniform elongation, the total elongation is only reduced by about 50% below 673 K. Post-irradiation strain-controlled fatigue tests have been carried out at $T_{\text{irrad}} = T_{\text{test}} = 523, 623$ and 723 K. Pronounced cyclic softening was observed in all specimens. At 623 and 723 K, neutron irradiation had no effect on fatigue life within the data scatter. A significant lifetime increase has been observed at $T_{\text{irrad}} = T_{\text{test}} = 523$ K that advances with decreasing stress amplitude $\Delta\epsilon$ (1% → 0.5%) up to a factor of ten. Scanning electron microscopy (SEM) analysis revealed ductile fracture and fatigue striations on the fracture surfaces. After push–pull fatigue testing, transmission electron microscopy (TEM) investigations showed the typical sub-cell formation, even at $T_{\text{irrad}} = T_{\text{test}} = 523$ K.

© 2013 Elsevier B.V. All rights reserved.

1. Introduction

The loading of first-wall and blanket structures of Tokamak-type fusion reactors, includes in addition to the plasma surface interactions, a combined effect of oscillating mechanical load and temperature gradients, which give rise to thermo-mechanical and thermal fatigue and also damage due to high fluxes of fusion neutrons. Therefore, for lifetime predictions and for the design of structural components, it is important to quantify the irradiation-induced changes on material properties such as low cycle fatigue (LCF) and tensile behavior. In the framework of the European steel developing program, the reduced activation ferritic/martensitic (RAFM) steel EUROFER97 [1–4] was irradiated up to 16.3 dpa in the mixed-spectrum reactor HFR of Petten, The Netherlands. A comprehensive post-irradiation program was performed, which included tensile, impact [5] and LCF tests. This paper is focused on the presentation of the tensile and LCF properties before and after irradiation. Due to the coolant incoming temperature at and below ~573 K of a blanket module, it is important to draw attention to the well-known irradiation-induced hardening and the related modification of mechanical properties. As the irradiation temperature ranged from 523–723 K, it covered the full range from pro-

nounced hardening at 573 K and below, up to minor irradiation-induced softening above ~673 K.

After irradiation, tensile tests and strain-controlled isothermal fatigue experiments were performed at $T_{\text{test}} = T_{\text{irrad}}$ and the fractographic and microstructural changes induced by irradiation and fatigue-testing were investigated by SEM and TEM. The tensile results are discussed and compared with earlier investigations done on RAFM steel irradiated at smaller damage doses [6,7] and the fatigue results are discussed in the context of recent post-irradiation examination results from fast-breeder-irradiated EUROFER97 at ~603 K [8].

2. Experimental procedure

Tensile and LCF specimens were fabricated of RAFM steel EUROFER97 (heat 83697); a 9%Cr–WVTa steel. The main elements of the alloy are 8.91 wt.% Cr, 0.12 wt.% C, 1.08 wt.% W, 0.48 wt.% Mn, 0.2 wt.% V, 0.14 wt.% Ta and 0.001 wt.% B. The material was tempered martensitic steel with typical lath structures in between prior austenitic grains with diameters of 18–24 μm . The final heat treatment of the 25 mm thick plate was 1253 K for 0.5 h cooled in air and tempered at 1033 K for 1.5 h. The hardness was 214–226 HV0.1 (Vickers Hardness).

The tensile specimens used had a gauge volume of $\varnothing 3 \text{ mm} \times 18 \text{ mm}$. The LCF specimens were smaller with a gauge volume of $\varnothing 2 \text{ mm} \times 7.6 \text{ mm}$ due to the international small specimen

* Corresponding author. Tel.: +49 721 608 22162; fax: +49 721 608 24567.

E-mail addresses: edeltraud.materna-morris@kit.edu (E. Materna-Morris), anton.moeslang@kit.edu (A. Möslang), hans-christian.schneider@kit.edu (H.-C. Schneider).

test technology (SSTT) program [9–11]. All specimens were oriented parallel to the rolling direction.

The total set of specimens was inserted in special irradiation rigs of a wrapper for the neutron irradiation in the central part of the HFR core in Petten (The Netherlands). The cumulative neutron fluence $E > 0.1$ MeV was $22.85 \times 10^{25} \text{ m}^{-2}$, resulting in a calculated displacement damage dose of 16.3 dpa for stainless steel for 771 full power days (FPDs). The irradiation temperatures ranged from 523 to 723 K.

The post-irradiation examinations were performed in the hot cell facility of the Fusion Material Laboratory at KIT (Karlsruhe Institute of Technology). A universal testing machine equipped with a high-vacuum furnace was used. Fully instrumented tensile tests were carried out with a strain rate of 0.1 mm/min at $T_{\text{test}} = T_{\text{irrad}}$, the test temperatures were the same as the irradiation temperatures. The unirradiated specimens were in the as-received condition. LCF tests were also performed at $T_{\text{test}} = T_{\text{irrad}} = 523, 623$ and 723 K in high vacuum. During these tests, complete fatigue cycles with hysteresis loops were recorded, which allowed a detailed monitoring of the stress amplitude $\Delta\sigma(N)$, as a function of cycles and of the elastic and plastic strains, $\Delta\epsilon_{\text{el}}(N)$ and $\Delta\epsilon_{\text{pl}}(N)$, respectively. The fatigue tests were performed under symmetric push-pull conditions until specimen failure N_f . N_f is defined as 80% stress-loss of the first cycle. After tensile and fatigue testing, the fracture surface and the characteristic microstructure of selected specimens were investigated by SEM and TEM. All TEM specimens were cut transverse to the gauge length. All hardness measurements were performed on non-deformed material of impact specimens, which were in the same irradiation program.

3. Results and discussion

3.1. Tensile properties

Figs. 1–3 show the tensile properties of irradiated and non-irradiated specimens. As expected, a significant irradiation-induced strength increase $\Delta\sigma_{\text{irr}}$, was observed at the lowest irradiation temperatures of 523 and 573 K, whereas above about 673 K a minor softening appears. This temperature behavior is expected and is a general feature of ferritic–martensitic steels, which is due to smaller dislocation loops and nanoscaled precipitates that are stable only below about 673 K [12]. As they are not formed above about 673–693 K, at least in a helium-free bcc matrix, uniform and total elongation are nearly the same at 723 K for irradiated and unirradiated specimens.

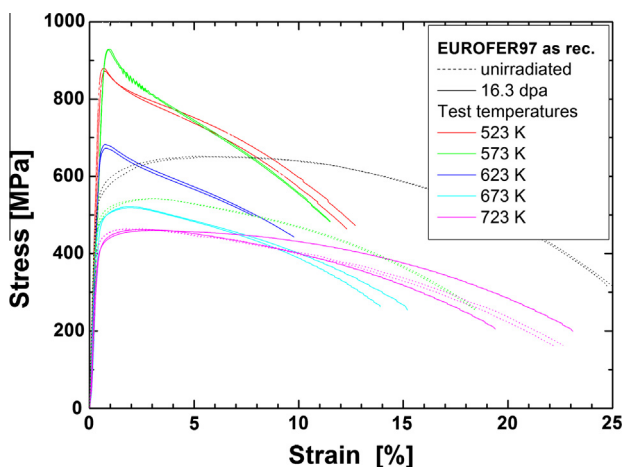


Fig. 1. Stress/strain diagram of EUROFER97.

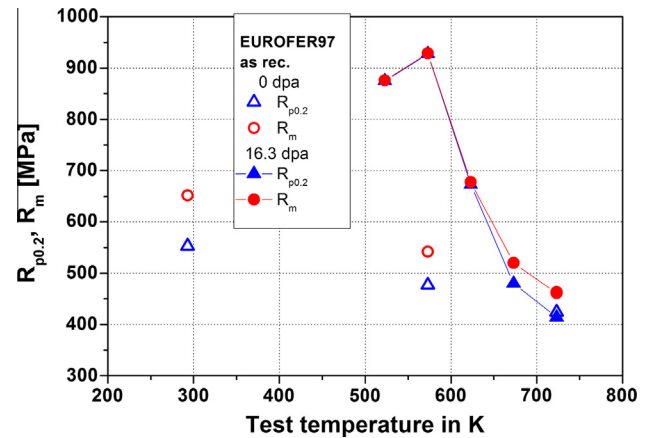


Fig. 2. Yield strength $R_{p0.2}$ and tensile strength R_m vs. test temperature.

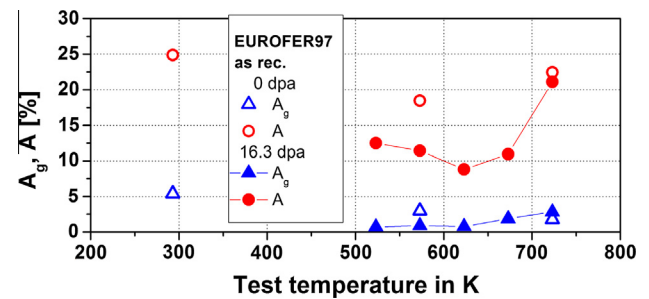


Fig. 3. Uniform elongation A_g and elongation after fracture A vs. test temperature.

However, irradiation at 523 K reveals a smaller $\Delta\sigma_{\text{irr}}$ of about 60 MPa compared with irradiation at 573 K. This observation might be surprising but is in agreement with hardness measurements that have been done after the same irradiation on F82H-mod and different EUROFER97 heats [13]. A general feature of mixed-spectrum neutron-irradiated ferritic–martensitic steels is also the serrated yielding or dynamic strain aging, which is observed at the selected strain rate around $T_{\text{test}} = T_{\text{irrad}} = 523$ K (Fig. 1). This is due to the repeated interaction between moving dislocations and solute carbon atoms. Despite the pronounced strain localization observed after irradiation below about 673 K and the related small uniform elongation A_g , of about 1% or even less, the total elongation after fracture A , is in the range of 10% and therefore, still about 50% of the unirradiated value. Consequently, low-temperature irradiated RAFM steels begin to deform around 1% in an irreversible manner by strain localization. Further analysis is required to determine whether dislocation channeling, observed in [14] as the origin of the strain localization, is also the dominant deformation mode under the present conditions.

3.2. LCF properties

The stress–strain hysteresis curves at 523 K revealed a distinct irradiation-hardening corresponding to the tensile tests [12]. An increase in strength of about 70% could be detected at 523 K and of about 60% at 623 K, whereas at 723 K, the stress amplitudes of the irradiated and the unirradiated specimens are almost identical, as expected from the tensile tests. In all diagrams of Figs. 4–6, the gradient in the stress amplitudes showed the typical cyclic softening of the RAFM steels. It is important to note that this cyclic softening is most pronounced at high strength, i.e. at low irradiation temperatures: for example, after 523 K irradiation and at a strain amplitude of 0.7% the stress amplitude collapses continuously

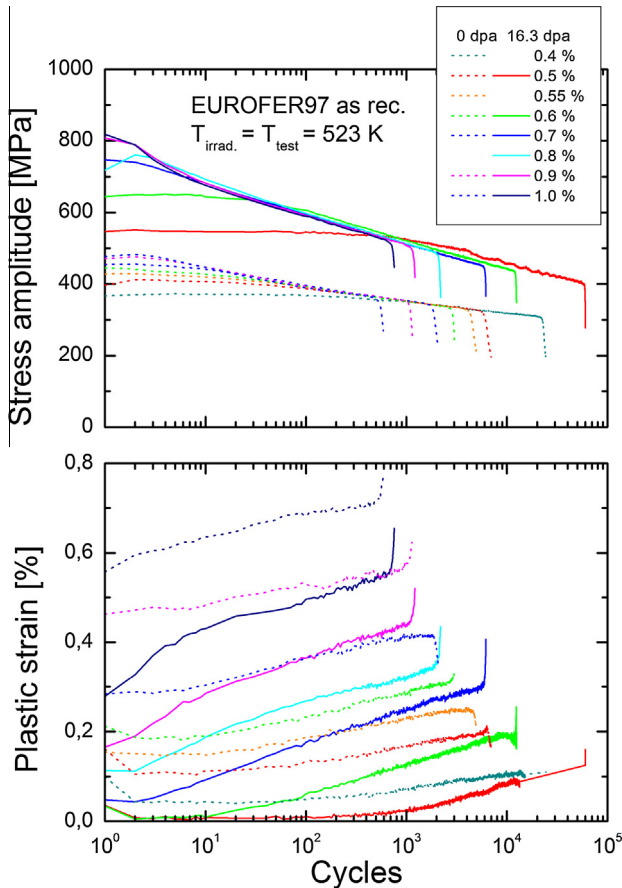


Fig. 4. Total stress amplitudes and plastic strains vs. cycles N during strain-controlled LCF tests for different strain ranges $\Delta\epsilon$ at 523 K.

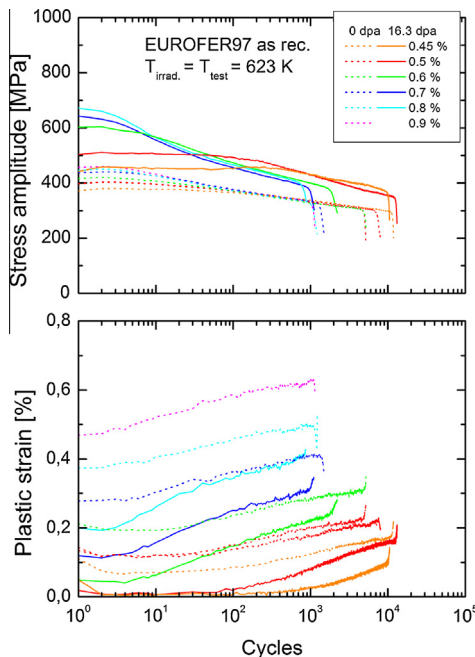


Fig. 5. Total stress amplitudes and plastic strains vs. cycles N during strain-controlled LCF tests for different strain ranges $\Delta\epsilon$ at 623 K.

from about 750 MPa during the first cycles down to about 500 MPa. Despite this cyclic softening, the stress amplitude of the irradiated specimens always remains significantly above that

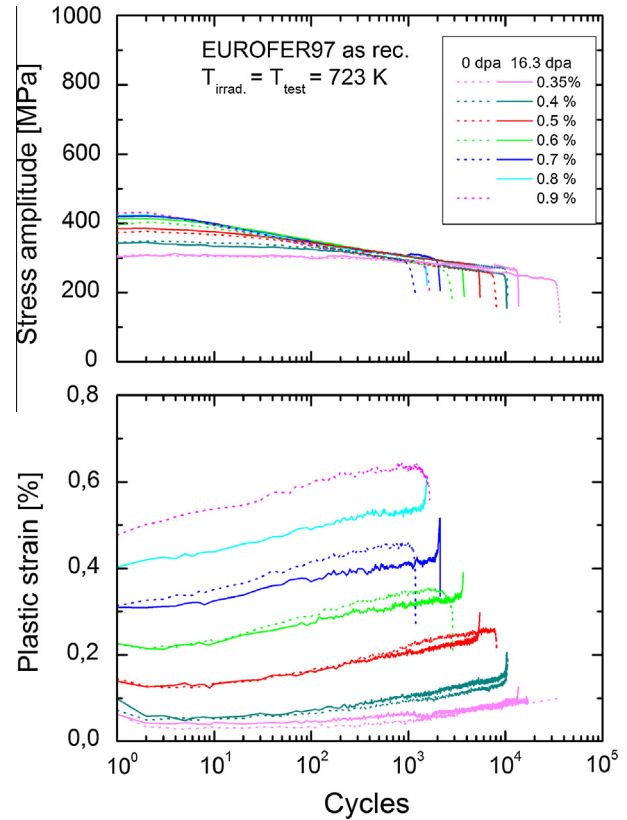


Fig. 6. Total stress amplitudes and plastic strains vs. cycles N during strain-controlled LCF tests for different strain ranges $\Delta\epsilon$ at 723 K.

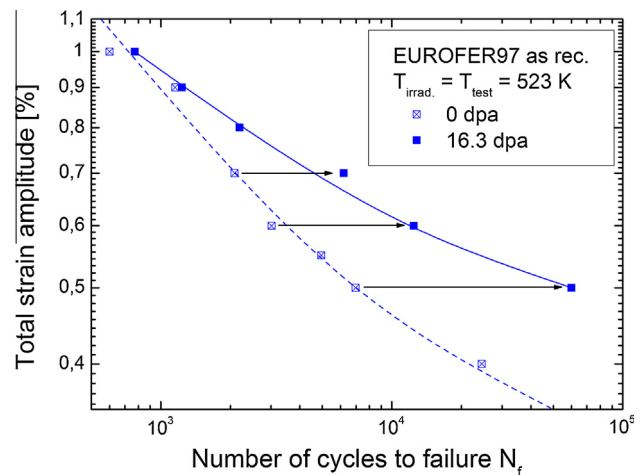


Fig. 7. Effect of 16.3 dpa irradiation on fatigue lifetime N_f at 523 K.

of the unirradiated specimens at irradiation temperatures below about 673 K. From this behavior it can be concluded that even the continuous cyclic motion of dislocations does not yet sweep out the vast majority of the irradiation-induced defects on their gliding planes.

Fig. 4 shows another common feature of LCF testing, which so far has only been observed in unirradiated specimens [15], all stress amplitudes follow one single common curve and are practically identical beyond about 200 cycles, despite the wide range of applied strain amplitudes. Consequently, large strain amplitudes show more cyclic softening at small strain ranges, i.e. the deformation response on individual fatigue loading occurs specifically in the early phase of fatigue testing.

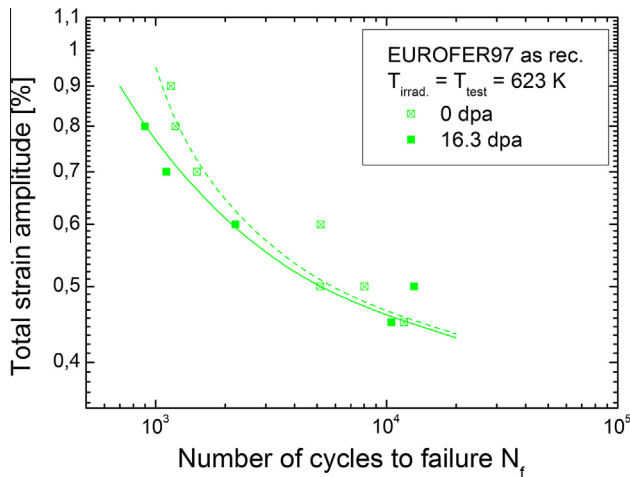


Fig. 8. Effect of 16.3 dpa irradiation on fatigue lifetime N_f at 623 K.

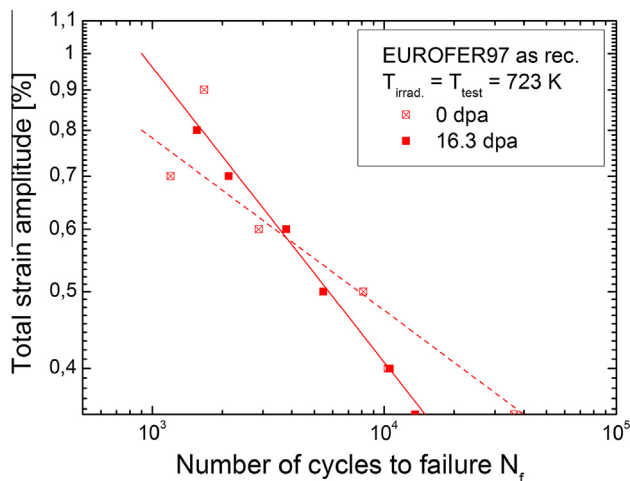


Fig. 9. Effect of 16.3 dpa irradiation on fatigue lifetime N_f at 723 K.

Also remarkable for the 523 K irradiation is that with decreasing strain amplitude the irradiated specimens achieved a considerable increase in fatigue life (Fig. 7). At the lowest $\Delta\epsilon$ of 0.5%, the number of cycles to failure N_f is about ten times higher than that in the unirradiated control. In contrast to the 523 K irradiations, the irradiation hardening is much less pronounced after 623 K irradiation (Fig. 8). This might be the reason that, for this irradiation temper-

ature, no clear irradiation effect on fatigue life could be determined within the data scatter. Also for the 723 K irradiation, the irradiated and unirradiated controls have a similar fatigue life within the experimental uncertainties, despite the somewhat different slopes in Fig. 9. The fatigue life results reported here on the 623 K irradiation are consistent with the ones reported on the same material after fast-breeder irradiation of up to 71 dpa at about 603 K [8].

3.3. Structural and fractographic properties

The irradiation-induced hardening could be quantified by Vickers hardness measurements. The steel had an initial hardness of 214–226 HV0.1 in the unirradiated state. As expected, a high increase was measured at 523 K and 16.3 dpa with 308–328 HV0.1 and also at 573 K with 321–356 HV0.1. At 623 K the hardness was lower again, between 265–282 HV0.1, and at 723 K, the thermal effect on the material could be determined with 221–237 HV0.1.

A homogeneous distribution of dislocation loops, point defects and/or α' -precipitates of about $2 \times 10^{15} / \text{cm}^{-3}$ was determined in EUROFER97 at 523 K and 16.3 dpa. These irradiation-induced microstructures are significantly responsible for the irradiation hardening. Some helium bubbles or voids having diameters of 1–2 nm were observed but their density was insufficient to influence the properties of the material.

After tensile tests at 523 K, an orientation mechanism of the dislocation loops and point defects was often observed [12]. A comparison with tensile-tested material at 723 K and 16.3 dpa showed a completely recovered material without any irradiation-induced loops or point defects but with a homogeneous sub-grain formation dependant on the thermal influence. Further microstructural changes were found after LCF tests. The thin foils for TEM were cut near the fracture. Sub-grain or sub-cell formation was found to be independent of the former microstructure with lath structure, as other laboratories have found [16–20]. The sub-cells in the unirradiated state were nearly free of dislocations. There was a clear difference to the irradiated material. In addition to dislocation-free sub-cells, sub-cells with a high density of irradiation induced defects and dislocation loops were observed. This development was seen especially after 523 K irradiation (Fig. 10a) but much less after 623 K. Following irradiation at 723 K (Fig. 10b), the LCF-tested specimens were practically free of dislocation loops. Independent of the irradiation temperature, TEM analysis of the irradiated specimens also revealed pronounced sub-cell formation similar to the unirradiated controls, i.e. significant fraction of dislocations move to the lath and grain boundaries,

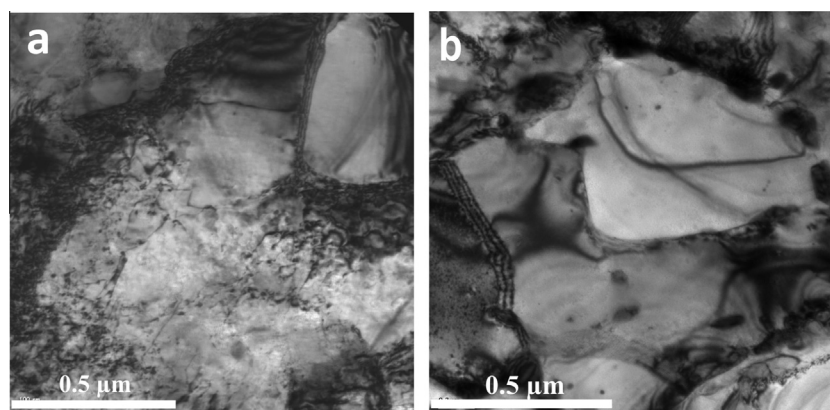


Fig. 10. Cell formation in LCF-tested specimens after 16.3 dpa at 523 K (a) and after 16.3 dpa at 723 K (b).

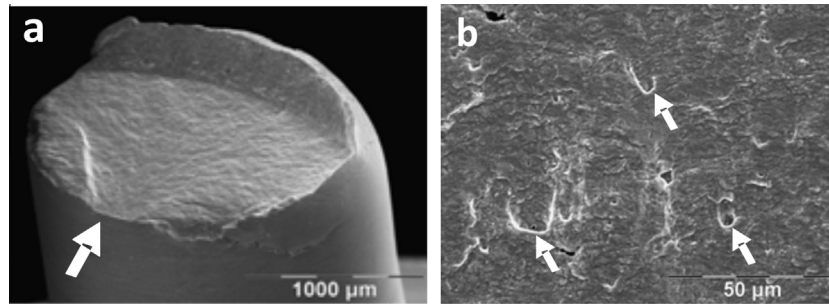


Fig. 11. Fracture surface of post-irradiated LCF tests at 723 K. (a) Shows the macroscopic view and the arrow marks the fracture origin. Details of the fracture are presented in (b) striation formation with cyclic formed dimples around small inclusions, shown by arrows.

leaving behind a cell interior with low dislocation density. This is a major reason for the cyclic softening observed during fatigue testing.

The fractures of the tensile specimens all had a ductile character with uniformly distributed dimple formations. The difference between the unirradiated and irradiated samples was quite evident. The irradiated samples showed less reduction of area compared with the unirradiated samples. The samples after LCF tests exhibited the typical flat fracture surface with striations which reflect the push–pull load on the material. Crack propagation is observable by fatigue paths and striations. In this way, the fracture origin of every LCF sample was clearly detectable at the outer face, as shown in Fig. 11. The first cracks are always initiated at the specimen's surface and depends on the highest stress state applied at the surface. Further surfaces are found in the bulk material, between, e.g. inclusions and the matrix. The cyclic load influenced these interfaces and large dimples with striations occurred around the inclusions (Fig. 11b).

4. Summary and conclusions

Irradiation-induced hardening, strength increase and related ductility reduction of EUROFER97 could be confirmed at irradiation temperatures <623 K. All tensile tests were performed at their irradiation temperatures. The highest tensile strength was observed at $T_{\text{irrad}} = T_{\text{test}} = 573$ K. Microstructural changes, such as dislocation loops, small precipitates and defects were observed due to the neutron irradiation. Developed dislocation structures are practically immobile at low irradiation temperatures and they caused the increase of hardness and strength.

At 623 K and higher, the density of irradiation-induced defects decreased by the thermal effect. A fully recovered microstructure with sub-grain formation was observed at 723 K. The mechanical properties achieved the level of unirradiated material.

Additionally, similar results for fatigue tests at higher irradiation and test temperatures could be attained. The lifetime after isothermal push–pull conditions at 623–723 K were in the range of the unirradiated specimens.

Special attention is due to the increase of lifetime at lower irradiation temperatures. EUROFER97 showed, after neutron

irradiation at 523 K, a remarkable increase of lifetime up to a factor of ten at $\Delta\epsilon = 0.5\%$.

In all LCF specimens of EUROFER97, the typical cyclic softening of 9%Cr-WVTa steels was observed in the unirradiated and irradiated conditions, reflected by the continuous decrease of stress amplitudes.

Sub-cell structures were already formed in all LCF specimens after a few 100 cycles. The irradiated microstructure still exhibited a high density of irradiation-induced defects. The number of recovered dislocation cells increases with increasing temperature. At 723 K the cell structures are predominantly recovered.

Acknowledgements

This work, supported by the European Communities under the contract of Association between EURATOM and KIT (Karlsruhe Institute of Technology), was carried out within the framework of the European Fusion Development Agreement.

References

- [1] K. Ehrlich, Philos. Trans. R. Soc. London A 357 (1999) 595.
- [2] N. Baluc et al., J. Nucl. Mater. 367–370 (2007) 33.
- [3] E. Lucon et al., Fusion Eng. Des. 81 (8–14) (2006) 917–923.
- [4] A. Möslang et al., Int. J. Mater. Res. 99 (2008) 1045.
- [5] E. Gaganidze et al., J. Nucl. Mater. 386–388 (2009) 349–352.
- [6] E. Lucon et al., J. Nucl. Mater. 329–333 (2004) 1078–1082.
- [7] T. Yamamoto et al., J. Nucl. Mater. 356 (2006) 27–49.
- [8] E. Gaganidze, C. Petersen, Post-irradiation examination of RAFM steels after fast reactor irradiation up to 71 dpa at <340 K (ARBOR 2), KIT Scientific Report 7596, 2011, ISSN: 1869–9669.
- [9] P. Jung et al., J. Nucl. Mater. 232 (1996) 186.
- [10] T. Hirose et al., J. Nucl. Mater. 283–287 (2000) 1018–1022.
- [11] E. Lucas et al., J. Nucl. Mater. 307–311 (2002) 1600.
- [12] E. Materna-Morris, A. Möslang, R. Rolli, H.-C. Schneider, J. Nucl. Mater. 386–388 (2009) 422–425.
- [13] I. Sacksteder, H.-C. Schneider, E. Materna-Morris, J. Nucl. Mater. 417 (2011) 127–130.
- [14] N. Hashimoto et al., Fusion Sci. Technol. 44 (2003) 490–494.
- [15] C. Vorpahl, A. Möslang, M. Rieth, J. Nucl. Mater. 417 (2011) 16–19.
- [16] B. van der Schaaf, J. Nucl. Mater. 155–157 (1988) 156.
- [17] R. Lindau et al., J. Nucl. Mater. 212–215 (1994) 599.
- [18] J. Bertsch et al., J. Nucl. Mater. 283–287 (2000) 832.
- [19] P. Marmy et al., J. Nucl. Mater. 377 (2008) 52.
- [20] S.W. Kim et al., J. Nucl. Mater. 386–388 (2009) 529.